

AUTOMATED VEHICLE CLASSIFICATION & COUNTING

AVC&C | Group G25 | University of Bradford

RESEARCH & LITERATURE REVIEW

CDIO Conceive Phase | Technology Evaluation | Dataset Analysis

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Project	Automated Vehicle Classification & Counting (AVC&C)
Module	COS6032-E Industrial AI Project
CDIO Phase	Conceive — Sprints 1 & 2 (September–November 2025)
Group	G25 — BSc Applied Artificial Intelligence
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1. Introduction & Research Objectives

The Conceive phase of the AVC&C project (Sprints 1–2, September–November 2025) involved structured research into the current state of computer vision-based traffic monitoring, a comparative evaluation of available AI models, and an assessment of suitable training datasets for UK road conditions.

Bradford Council currently deploys 2–3 traffic survey officers per road junction to conduct manual vehicle counts. This approach is expensive (estimated £15.30/hour per officer), inconsistent across surveyors, and provides only snapshot data rather than continuous monitoring. The research phase sought to establish whether a proven, open-source computer vision solution could replace this process at a cost accessible to a local authority.

This document presents the research findings that informed the Group G25 technical architecture decisions, dataset strategy, and model selection.

2. Problem Domain Research

2.1 Traffic Monitoring in Smart Cities

Automated traffic monitoring is a well-established computer vision application, with academic and commercial systems deployed globally. The UK Department for Transport (DfT) notes that accurate traffic data is essential for Local Transport Plans, active travel infrastructure investment, and road safety analysis. However, automated systems have historically been cost-prohibitive for smaller local authorities, with proprietary ANPR and traffic counting systems costing £10,000–£50,000+ per installation.

Recent advances in real-time object detection — particularly the YOLO (You Only Look Once) architecture family — have reduced the computational requirements for accurate vehicle detection to the point where CPU-only deployment on commodity hardware is viable. This created the opportunity for an open-source, low-cost alternative suitable for Bradford Council's budget and technical capacity.

2.2 Bradford Council Context

- **Current method:** Manual surveys: 2–3 officers per junction, paper tally sheets, snapshot data only
- **Data need:** Vehicle counts by type (car/bus/truck) for Local Transport Plan evidence base
- **Budget constraint:** No dedicated hardware budget — must run on existing Council laptops
- **Privacy requirement:** No personal data capture — GDPR compliance non-negotiable
- **Weather requirement:** Bradford's climate requires robustness to rain, fog, and snow

3. Object Detection Model Comparison

Group G25 conducted a structured evaluation of candidate real-time object detection models during Sprint 1. The evaluation criteria were derived directly from Bradford Council's requirements: accuracy, CPU speed, open-source availability, and ease of deployment.

Model	mAP@0.5	CPU FPS	Licence	G25 Assessment
YOLOv8n	~87%	18–22 FPS	AGPL-3.0	Strong — established ecosystem, good docs
YOLOv9n	~88%	15–18 FPS	GPL-3.0	Slightly slower on CPU — less tooling
YOLO11n	~91%+	25+ FPS	AGPL-3.0	SELECTED — best accuracy + CPU speed
Faster R-CNN	~89%	5–8 FPS	Apache 2.0	Too slow for real-time on CPU
SSD MobileNet	~78%	28–35 FPS	Apache 2.0	Fast but below 85% accuracy target
RT-DETR	~91%	8–12 FPS	Apache 2.0	Accurate but too slow on CPU

Rationale for YOLO11n Selection

- **Accuracy:** mAP@0.5 exceeds the 85% Bradford Council target on COCO benchmark
- **CPU Speed:** 25+ FPS on AMD Ryzen 5 5600H — meets real-time requirement without GPU
- **Ecosystem:** Ultralytics library provides simple Python API, pretrained weights, and active community
- **COCO Pretrained:** 80-class COCO pretraining includes cars (class 2), buses (class 5), and trucks (class 7) — directly applicable to Bradford Council's vehicle types
- **Lightweight:** Nano variant (YOLO11n) — smallest model size, lowest memory footprint, fastest CPU inference

4. Multi-Object Tracking Research

Vehicle counting requires not just detecting vehicles in individual frames but tracking them across frames to avoid duplicate counts. Group G25 evaluated tracking algorithms during Sprint 3.

Algorithm	Approach	CPU Performance	G25 Decision
DeepSORT	Kalman filter + deep appearance features	Low overhead — suitable for CPU	SELECTED

BoTSORT	Improved re-identification, camera motion	Slightly higher overhead	Considered — DeepSORT simpler
ByteTrack	High-performance, occlusion handling	Moderate CPU overhead	Evaluated — overkill for our use case
SORT (Simple)	IoU-based matching only	Very fast	Rejected — poor re-ID after occlusion

DeepSORT was selected for its balance of tracking accuracy and CPU efficiency. The persistent track ID per vehicle enables the line-crossing counting logic in `vehicle_counter.py` — each unique vehicle ID is counted exactly once as it crosses the virtual trip-line at $y=300$.

5. Dataset Research & Selection

5.1 Publicly Available Traffic Datasets

Group G25 researched publicly available traffic datasets during Sprint 2. The key requirement was weather diversity — specifically rain, fog, and snow conditions representative of Bradford's climate.

Dataset	Size	Weather Variety	G25 Assessment
UA-DETRAC	140,000+ frames	Some variation	Beijing-based — excellent for benchmarking, not UK training
COCO (vehicles)	118,000 images	Mostly clear	Good pretrain base — lacks UK road context and weather
KITTI	~15,000 frames	Limited	German roads — not directly applicable to UK
CityScapes	~5,000 images	Limited	Urban segmentation focus — annotation format unsuitable
Roboflow Universe	Variable	Configurable	SELECTED — weather-specific collections available

5.2 Custom Dataset Construction (Roboflow)

No single public dataset met all requirements. Group G25 constructed a custom training dataset using Roboflow, combining and annotating images across five weather conditions to create a UK-applicable, weather-robust training set.

Weather Condition	Images	Source	Notes
Normal	~2,400	Roboflow Universe	Clear conditions — baseline performance
Rain	~2,300	Roboflow Universe	Wet roads, spray, reduced contrast
Fog	~2,200	Roboflow Universe	Low visibility — most challenging condition
Snow	~2,300	Roboflow Universe	White road surfaces reduce vehicle contrast
Sand/Haze	~2,382	Roboflow Universe	Included to improve fog-like condition robustness
TOTAL	11,582	Custom compiled	70% train / 22% val / 15% test split via prepare_dataset.py

Dataset labelling was led by Pratham Patel, requiring approximately 25 hours of manual annotation effort. All annotations were performed using Roboflow's web interface and exported

in YOLO format (data.yaml + class labels). The prepare_dataset.py script written by Ayush Acharya automates the 70/22/15% split and directory organisation.

6. Dashboard & Visualisation Research

Bradford Council requires outputs in a format accessible to non-technical traffic planning officers. Group G25 evaluated dashboard frameworks during Sprint 5.

Framework	Strengths	Weaknesses	Decision
Streamlit	Python-native, fast to build, Plotly integration, no JS required	Limited customisation vs React	SELECTED
Dash (Plotly)	More customisable, good for analytics	More complex setup, more boilerplate	Rejected — overkill for scope
Grafana	Powerful time-series, industry standard	Requires separate data source setup	Rejected — too complex for local deployment
Flask + Chart.js	Full control, lightweight	Requires HTML/JS development	Rejected — team strength is Python

Streamlit was selected as it enables rapid Python-native dashboard development with Plotly charts, requires no JavaScript, and can be launched locally with a single command — essential for Bradford Council officers who are not technically trained.

7. Related Work

7.1 Academic Context

The AVC&C system builds on a substantial body of academic and industry research in computer vision-based traffic monitoring. Key related works that informed Group G25's approach include:

- **Bochkovskiy et al. (2020):** YOLOv4: Optimal Speed and Accuracy of Object Detection. arXiv:2004.10934. Foundational work establishing YOLO as the standard for real-time object detection, directly informing our model selection rationale.
- **Wojke et al. (2017):** Simple Online and Realtime Tracking with a Deep Association Metric (DeepSORT). IEEE ICIP 2017. The tracking algorithm at the core of our counting pipeline.
- **Wen et al. (2020):** UA-DETRAC: A New Benchmark and Protocol for Multi-Object Detection and Tracking. Computer Vision and Image Understanding. The benchmark dataset used for our external validation.
- **Ultralytics (2024):** YOLO11 Technical Documentation. docs.ultralytics.com. Primary technical reference for YOLO11n training configuration and inference.
- **Wang et al. (2023):** YOLOv9: Learning What You Want to Learn Using Programmable Gradient Information. arXiv:2402.13616. Reviewed during model comparison — informed our YOLO11n selection.

7.2 Industry Context

- **Hikvision & Axis ANPR:** Commercial benchmark: typical local authority ANPR deployment costs £10,000–£50,000+ per junction. AVC&C delivers comparable counting capability at ~£15 (poster printing only).
- **Google's Traffic Layer:** Cloud-based traffic monitoring using aggregated mobile device data — privacy-invasive and subscription-based. AVC&C is privacy-preserving and zero ongoing cost.
- **Bradford Council Smart City Programme:** Bradford Council's 2023 Digital Strategy identifies automated monitoring as a key enabler for evidence-based transport planning. AVC&C directly addresses this strategic need.

8. Research Conclusions

The Conceive phase research established the following key conclusions that shaped the AVC&C technical architecture:

#	Research Conclusion	Architectural Decision
1	YOLO11n is the optimal model for CPU real-time detection	Selected as core detection engine — trained from COCO pretrained weights
2	No public dataset meets UK weather requirements	Custom 11,582-image dataset built on Roboflow across 5 weather conditions

3	DeepSORT provides best tracking accuracy/CPU balance	Integrated as tracking layer between detection and counting logic
4	Streamlit is the optimal dashboard for non-technical officers	app.py built in Streamlit with Plotly charts and weather filter
5	UA-DETRAC is the appropriate external benchmark	Used for independent validation in Sprint 7 — not for training
6	Privacy-by-design is essential from day 1	GDPR compliance built into CSV schema — no personal data at any stage

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